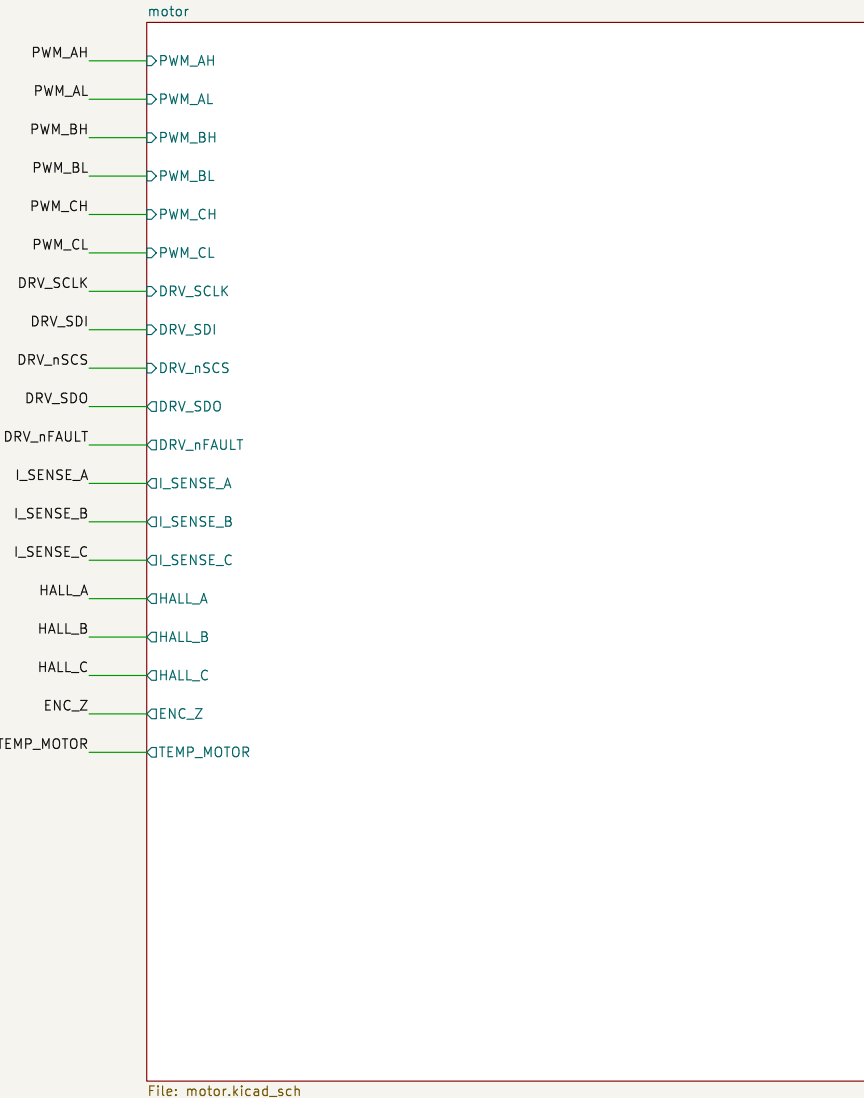
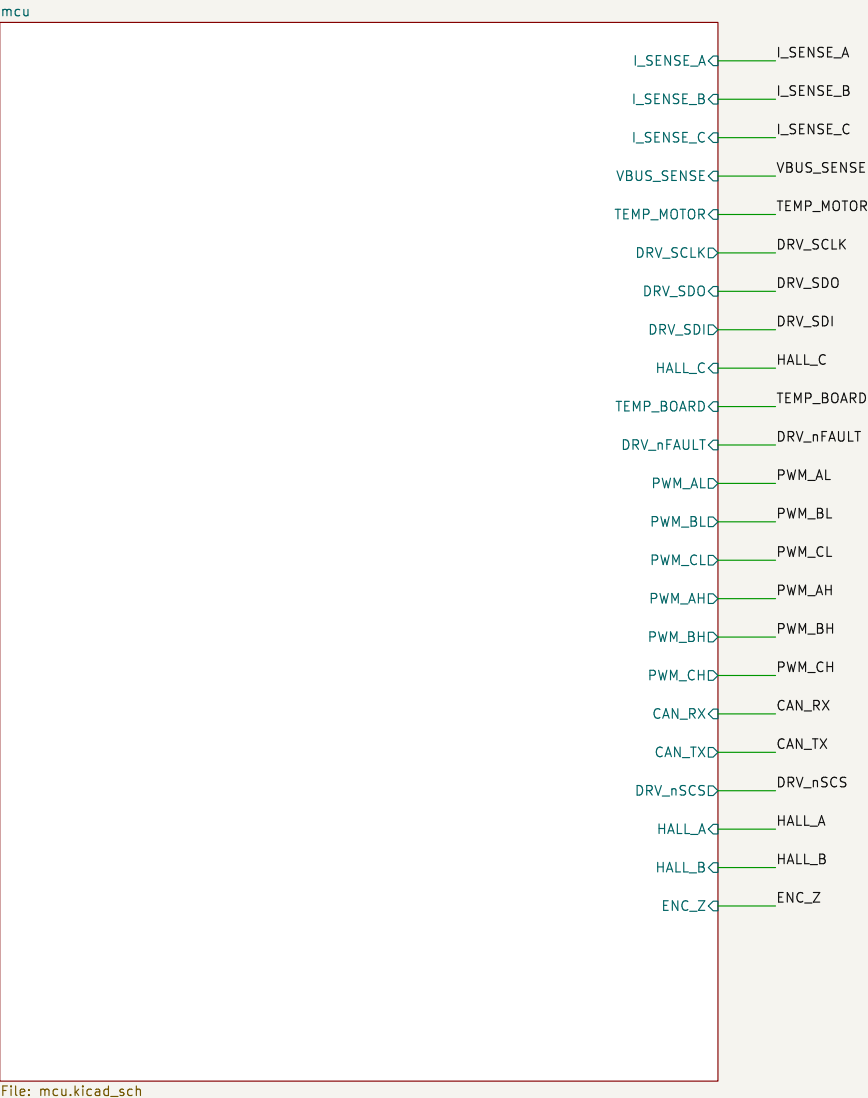


STM32G431 CAN-FD actuator node – one node per actuator



Bus: FDCAN1 at 1 Mbit/s arbitration, up to 5 Mbit/s data.

Rails: VM (9–18 V) → 5 V buck → 3.3 V LDO → filtered 3.3 V analog.

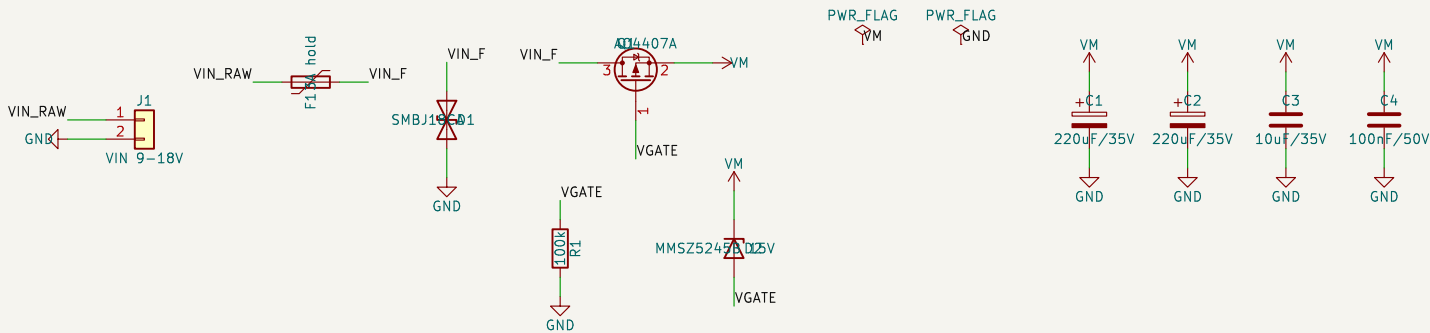
One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
Top level

Sheet: /
File: g431-can-actuator.kicad_sch

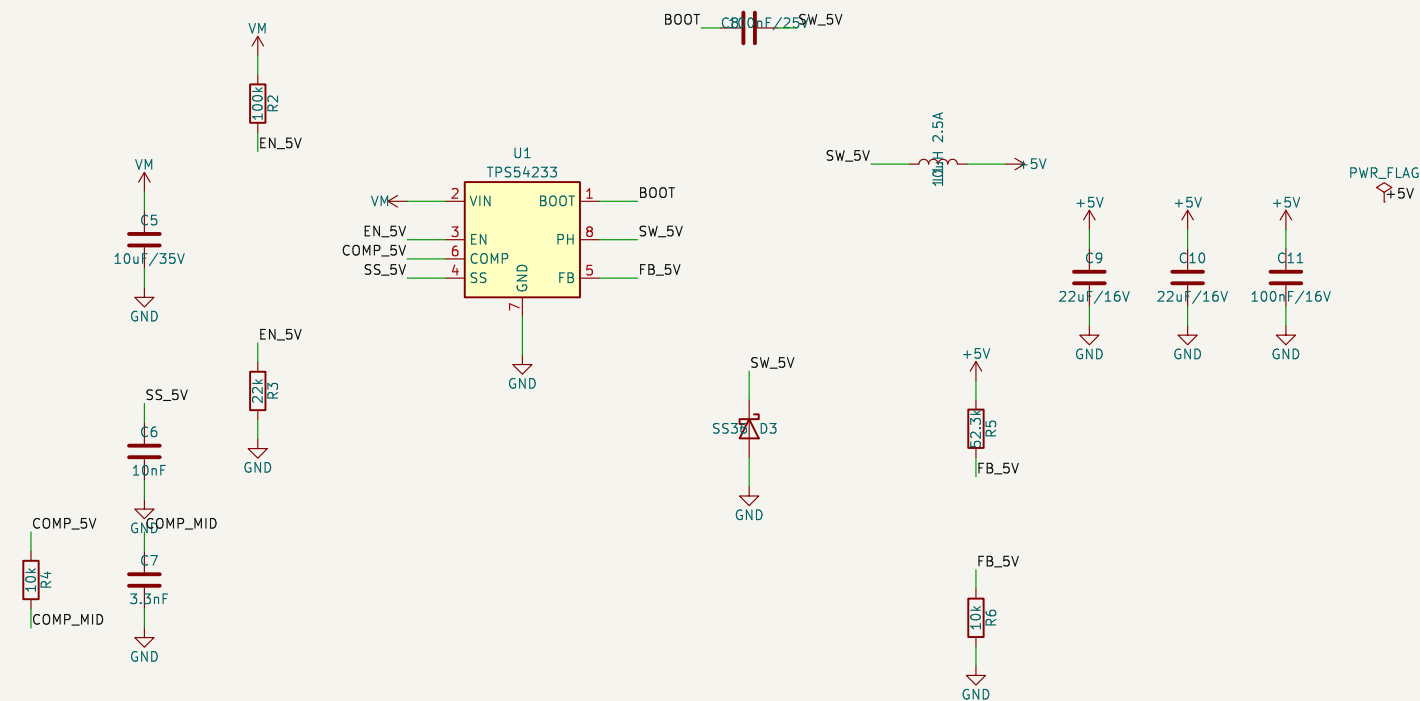
Title: STM32G431 CAN-FD actuator node

Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

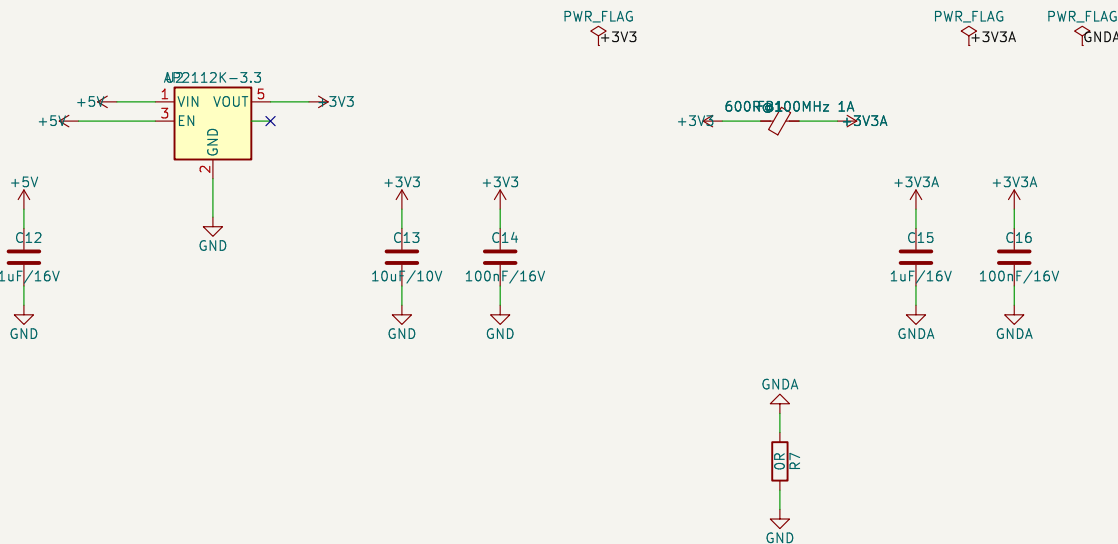
Input protection – 9..18 V DC (12 V nominal), 20 V absolute max



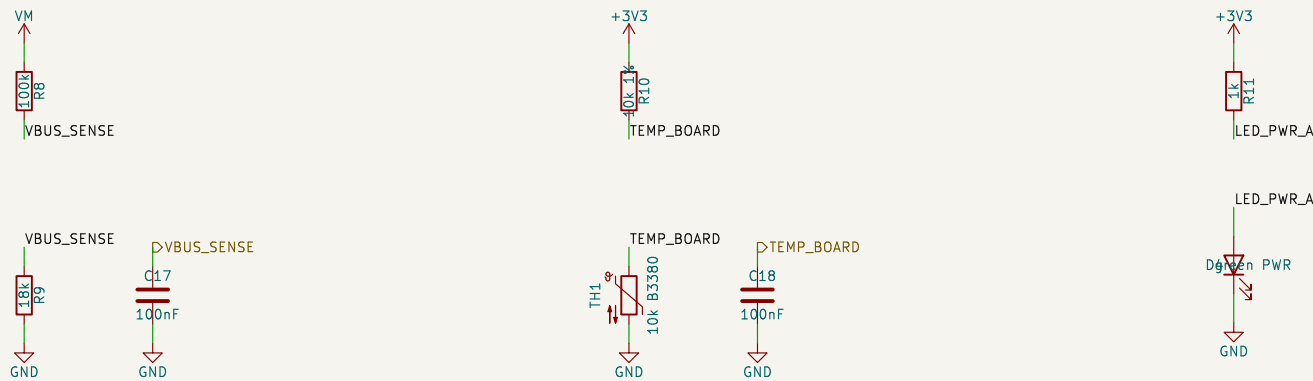
5 V / 2 A step-down – TPS54233, 570 kHz, non-synchronous



3.3 V logic rail and filtered analog rail



Bus voltage and board temperature sensing



One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
Power input, protection and rails

Sheet: /power/
File: power.kicad_sch

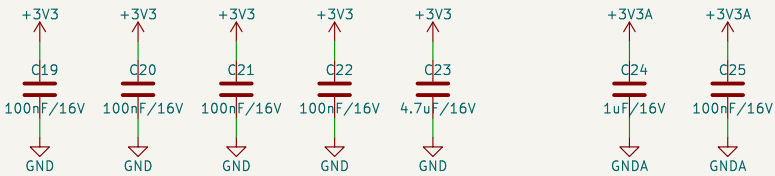
Title: STM32G431 CAN-FD actuator node

Size: A3 Date: KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

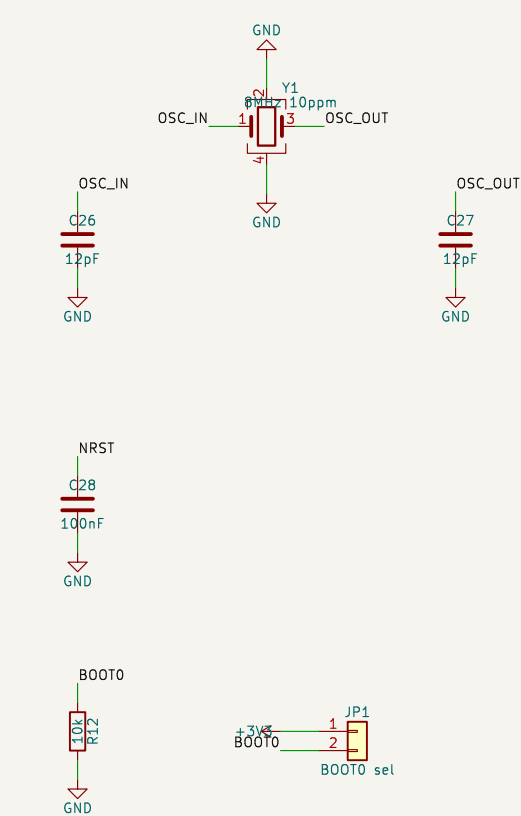
Rev: A

Id: 2/5

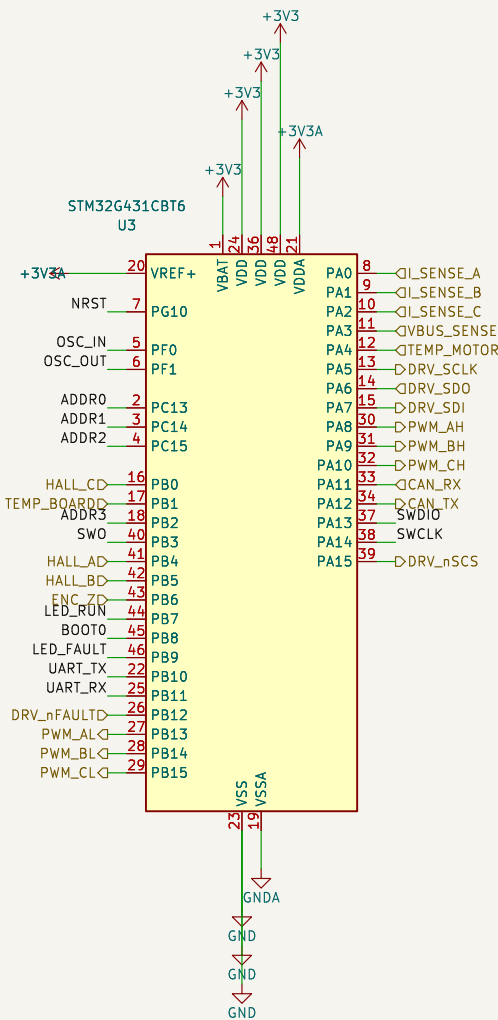
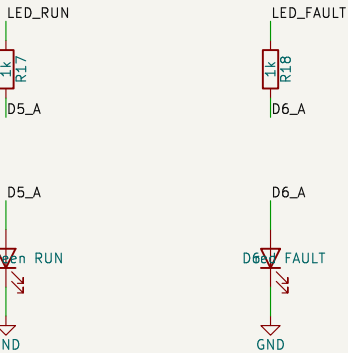
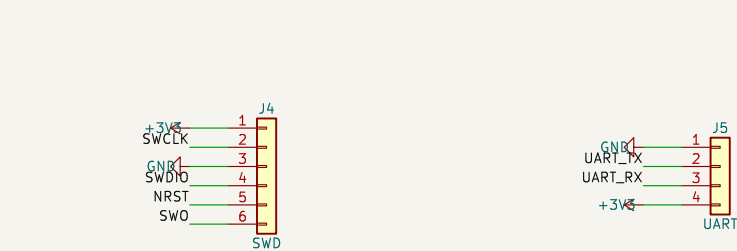
Decoupling: one 100 nF per supply pin, placed at the pin



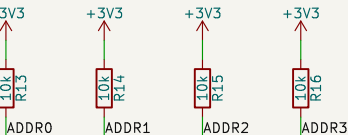
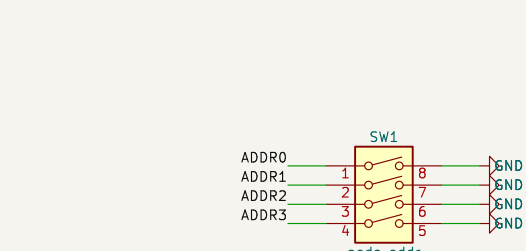
8 MHz HSE – PLL to 170 MHz, needed for CAN bit timing



SWD (Cortex–M pinout) and debug UART

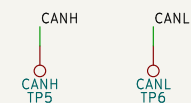
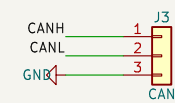
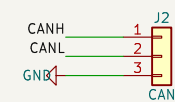
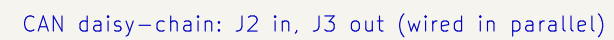
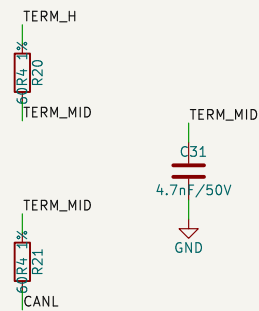
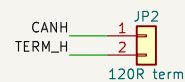
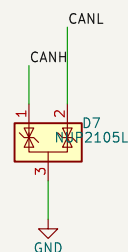
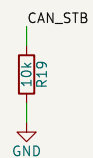
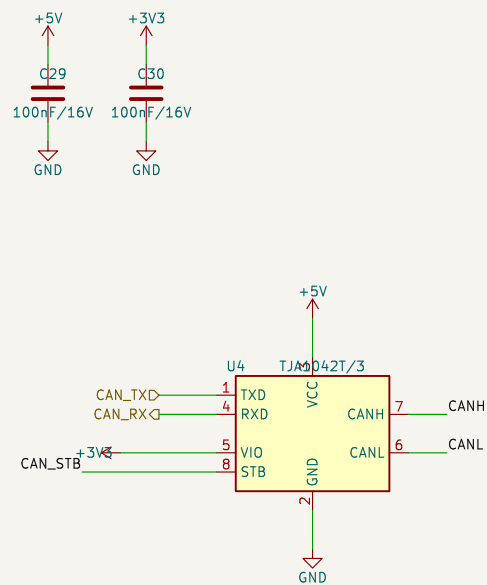


Node address – 16 actuators per bus; switch closed = 0



One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
STM32G431CBT6, clock, debug and I/O

Sheet: /mcu/ File: mcu.kicad_sch		
Title: STM32G431 CAN–FD actuator node		
Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		
Id: 3/5		



One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
CAN-FD transceiver and bus interface

Sheet: /can/
File: can.kicad_sch

Title: STM32G431 CAN-FD actuator node

Size: A3	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

Date:	Rev: A
cad-cli 7.0.11+dfsg-1build4	Id: 4/5

Rev: A

DRV8311S – integrated three-phase bridge, 3..20 V, 5 A peak,
three integrated current-sense amplifiers, SPI control

ADC input networks – charge reservoir for the SAR sampling
capacitor, $f_c \sim 1.6$ MHz (settles inside the sample window)

Feedback: Hall (TIM3 XOR) or quadrature encoder + motor NTC

One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
DRV8311S three-phase stage and feedback

Sheet: /motor/
File: motor.kicad_sch

Title: STM32G431 CAN-FD actuator node

Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5

DRV8311S – integrated three-phase bridge, 3..20 V, 5 A peak,
three integrated current-sense amplifiers, SPI control

ADC input networks – charge reservoir for the SAR sampling
capacitor, $f_c \sim 1.6$ MHz (settles inside the sample window)

Feedback: Hall (TIM3 XOR) or quadrature encoder + motor NTC

One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
DRV8311S three-phase stage and feedback

Sheet: /motor/
File: motor.kicad_sch

Title: STM32G431 CAN-FD actuator node

Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5

DRV8311S – integrated three-phase bridge, 3..20 V, 5 A peak,
three integrated current-sense amplifiers, SPI control

ADC input networks – charge reservoir for the SAR sampling
capacitor, $f_c \sim 1.6$ MHz (settles inside the sample window)

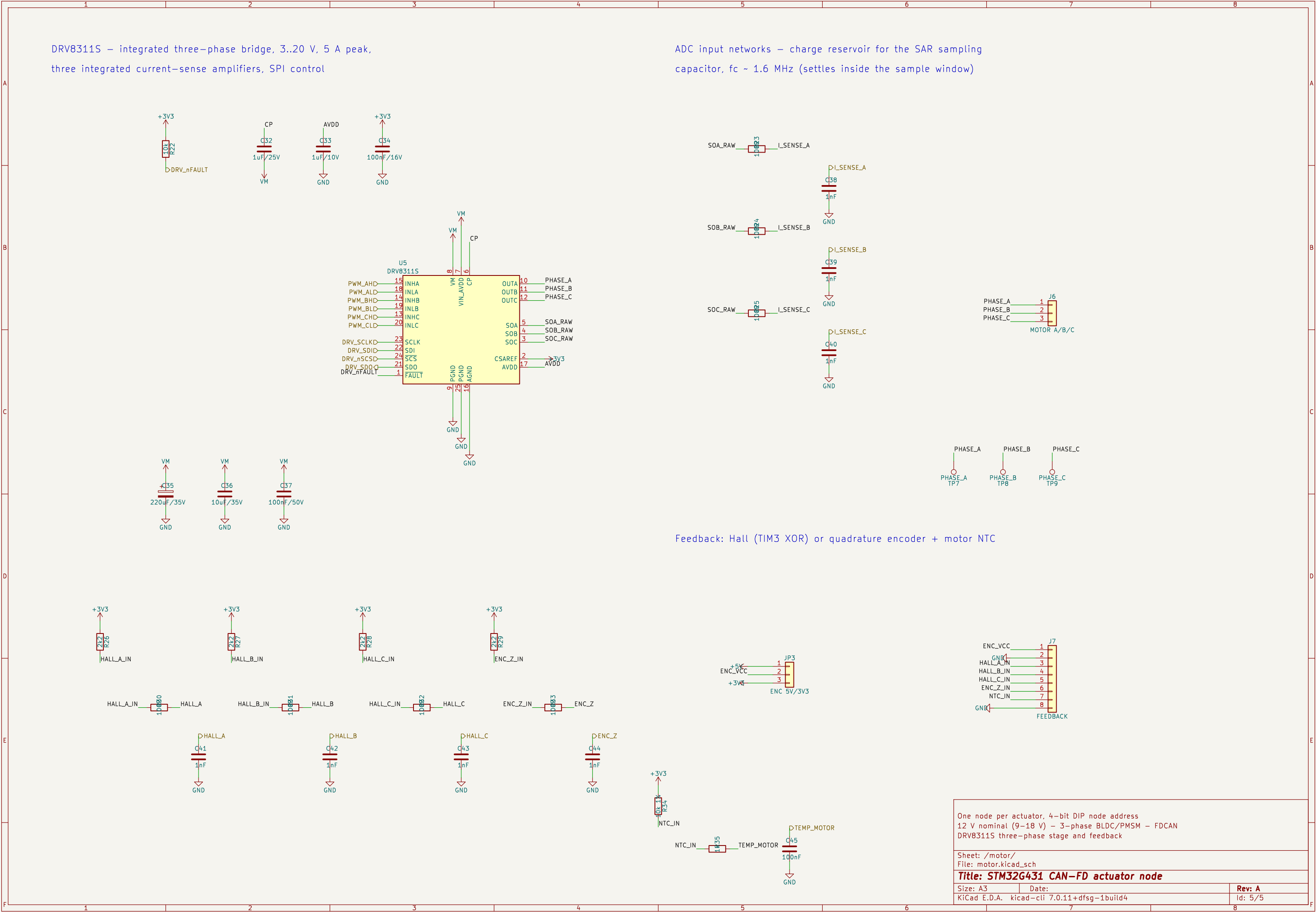
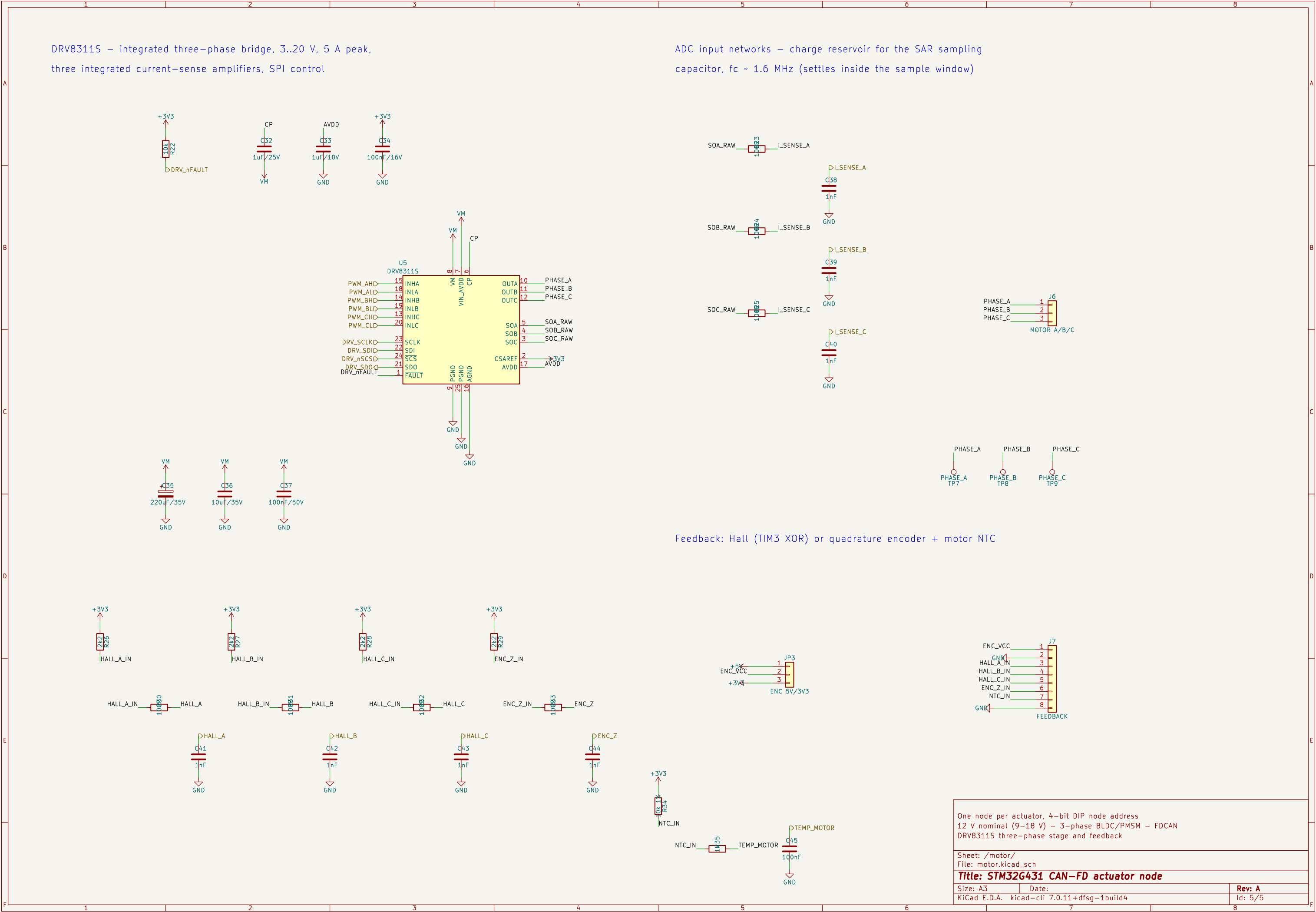
Feedback: Hall (TIM3 XOR) or quadrature encoder + motor NTC

One node per actuator, 4-bit DIP node address
12 V nominal (9–18 V) – 3-phase BLDC/PMSM – FDCAN
DRV8311S three-phase stage and feedback

Sheet: /motor/
File: motor.kicad_sch

Title: STM32G431 CAN-FD actuator node

Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5



Size: A3	Date:	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5